

```

public static int task3CP(int[] input) {
    int maxLength = 0; // Variable to store the length of the longest subarray
    int sum = 0; // Variable to keep track of the cumulative sum of elements
    HashMap<Integer, Integer> sumMap = new HashMap<>(); // Key: cumulative sum, Value: index of first occurrence
    for (int i = 0; i < input.length; i++) { // Iterate through the input array
        sum += input[i]; // if the same cumulative sum is encountered again, it means that
                        // the elements between the two occurrences have a sum of 0.
        if (sumMap.containsKey(sum)) { // There is a subarray with the desired sum.
            maxLength = Math.max(maxLength, i - sumMap.get(sum)); // Update the maximum length if necessary
        } else {
            sumMap.put(sum, i); // Otherwise, add the cumulative sum and its index to the map
        }
    }
    return maxLength; // Return the length of the longest subarray
}

```

```

public static void main(String[] args) {
    int[] input = {15, -2, 2, -8, 1, 7, 10, 23};
    int result = task3CP(input);
    System.out.println(result);
}

```